

Essential Data Analyst Course Curriculum

- **Introduction to Data Science & Analytics** (Basic Concepts)
- **Statistics:** Descriptive Statistics (Mean, Median, Mode, Variance, Std Dev)
- **Statistics:** Normal Distribution, Z-Score, and Central Limit Theorem
- **Statistics:** Hypothesis Testing, P-Values, and Type I/II Errors
- **SQL:** Basic Queries (SELECT, FROM, WHERE, GROUP BY)
- **SQL:** Advanced Joins (Left, Right, Inner, Full) and Unions
- **SQL:** Window Functions (RANK, ROW_NUMBER, LEAD, LAG)
- **SQL:** Common Table Expressions (CTEs) and Subqueries
- **Python:** NumPy for Numerical Data
- **Python:** Pandas DataFrames (Cleaning, Filtering, Grouping) - **High Priority**
- **Python:** Handling Missing Values and Data Wrangling
- **Data Visualization:** Matplotlib and Seaborn basics
- **Exploratory Data Analysis (EDA) Projects** - Complete at least 2
- **Power BI/Tableau:** Connecting Data and Building Dashboards
- **Power BI:** DAX Expressions for Calculations
- **Capstone Projects:** Real-world business case studies